Addressing Racism In Football With Natural Language Processing: Related Works And Research

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Bias Mitigation in the NLP field

One of the challenges of our project is addressing bias mitigation in NLP language models and its shortcomings. From the paper **Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science** by Emily M. Bender and Batya Friedman, we know that two of the biggest potential problems that occur from NLP models are the inability to work for specific subpopulations that are not supported by test or training data, as well as systems that reinforce the biases inherent in the training data. Because models are only built on what we feed, the latter is especially important to avoid reinforcing bias. One way to do this according to the paper is the use of data statements as a test of the content in the training data; the data statements will expose any potential bias or negative sentiments to researchers while also giving a summary of what group of topics the data has included [Bender and Friedman, 2018]. This allows the researchers to figure out if any key subpopulations are being addressed as well as get a good gauge of inherent biases in the data. There is importance to dataset transparency as well, which will allow future researchers the full set of tools to potentially tackle any unaddressed biases still left in the research.

In **A Survey of Race, Racism, and Anti-Racism in NLP,** Field and colleagues undergo a critical analysis of 79 papers within the ACL that address racial aspects in NLP. This paper exposes how race has been traditionally approached in the field, uncovering persistent research gaps and the oftentimes siloed treatment of race as a subject matter. To fully understand the problem of race as it pertains to NLP, it is necessary to engage with the definition of racism or the identification of racial bias, especially in hate speech classifiers. We must also move beyond the single label classifications to engage with the complex socio-historical context of the language itself. Historically, colonial powers constructed linguistic and racial hierarchies to justify violence, and currently, beliefs about the inferiority of racialized people's language practices continue to justify social and economic exclusion [Field et al., 2021]. Having a full context of the language given to our model, having varied ranges of classification labels (which we have set into place in our project plan), and using data statements to check the temperature of our given data are essential ways to mitigate bias, especially in terms of tackling racism in digital football spaces.

Racist Sentiments in Social Media Datasets

Muhammad Asif et al.'s paper, **"Sentiment Analysis of Extremism in Social Media from Textual Information,"** outlines methods relevant to our research goals by addressing the complex problem of identifying extremist content on social media. This is done by gathering data from Facebook and applying lexicon-based sentiment analysis to label it effectively. They further analyze this data with solid machine learning techniques, using tools like Multinomial Naïve Bayes and Support Vector Machines provided by the Scikit Python library [Asif et al., 2020].

Their approach is particularly applicable to our work on football forums where nuanced conversations can often conceal racist sentiments. By adopting a lexicon-based analysis, we can develop a system that's sensitive to the context and subtleties of football-related discussions. As we begin our research, the methods used by [Asif et al., 2020]. offer a practical framework that we can benchmark for our analysis. Using these methods will help us pinpoint and address the subtle racism that may be present in the lively debates and commentary within the football community on social media.

Detection of Subtle and Veiled Hate Speech

In the world of natural language processing, spotting hate speech that's not immediately obvious is a serious challenge that our upcoming research intends to tackle. This issue is at the heart of Han & Tsvetkov's paper **"Fortifying Toxic Speech Detectors Against Veiled Toxicity."** They zoom in on the difficulty of recognizing hate speech that is cleverly disguised. The researchers' solution doesn't rely on having a massive database of these subtle hateful comments. The paper details a method of enhancing a classifier's ability to detect veiled hate speech without creating a model from scratch. Instead, they enhance the model's detection capabilities by using influence functions to identify how training examples affect the classifier's ability to detect subtle toxicity [Han and Tsvetkov, 2020]. By fine-tuning a BERT model with carefully chosen examples of veiled toxicity, they address the challenge without the need for a large new dataset of labeled veiled toxic speech. This helps the model get better at spotting the kind of sneaky comments that traditional systems might miss.

For us, this method could be a game-changer. As we're about to dive into the intersection of hate speech and football, understanding the subtleties of how hate speech hides in plain sight can guide our initial steps. Football, with its passionate fans and heated debates, can often hide racist undertones in seemingly innocent comments. Han & Tsvetkov's approach gives us a valuable tool for our toolkit. By adapting their strategy, we can develop a model that not only looks for the obvious signs of racism but also reads between the lines, capturing the hidden biases that are just as harmful. As we set out to collect our own dataset from various football-related online discussions, their method points us toward a more nuanced understanding of language. It teaches us the ways to look for subtle signs of hate speech, especially the ones that might slip by unnoticed.

Challenges in Automated Sentiment Analysis for Hate Speech Detection

Automated sentiment analysis for hate speech detection is fraught with challenges, as discussed in **"Detecting Hate Speech from Tweets for Sentiment Analysis"** by Lin Jiang and Yoshimi Suzuki. While advances have been made, the accurate interpretation of sentiment in tweets—a common medium for football fan interaction—remains a complex task. The paper illustrates the difficulties in distinguishing between casual language and potentially harmful expressions [Jiang and Suzuki, 2019], a distinction critical in the real-time world of social media where fans often react instantaneously to events. Refining these automated tools is critical for monitoring and mitigating hate speech in the domain of sports, contributing to a respectful and inclusive community for all supporters.

References

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